

08

VIRTUAL MEMORY

8.1 VM Concept

- It gives an illusion to the programmer that programs having larger size than the physical memory can be executed.
- It allows address spaces to be shared by several processes.

8.2 VM Implementation

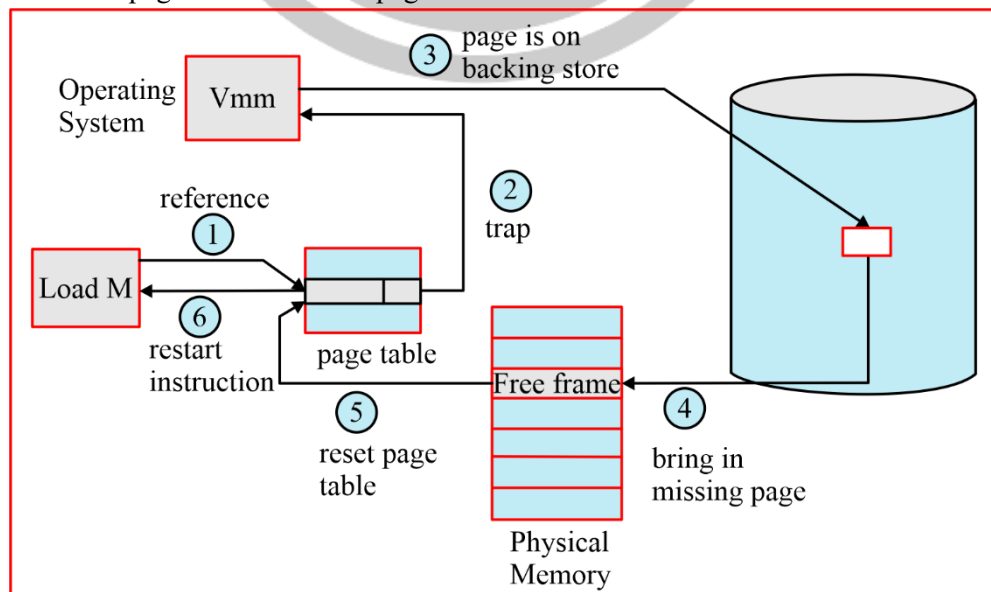
Virtual memory can be implemented through:

- Demand Paging
- Demand Segmentation

8.2.1 Demand Paging

Loading the pages from the secondary memory to the main memory on demand is called 'demand paging'.

- When the CPU tries to refer a page which is not available in the main memory, 'page fault' occurs. A signal is sent to the OS regarding the page fault. The OS locates the page into the Logical address space and loads it into the main memory frame. If the memory frames are full, appropriate page replacement algorithm is used. The respective page table entries are also updated accordingly.
- The time taken to service a 'page fault' is called 'page fault service time'.



8.3 Performance of virtual Memory

8.3.1 Temporal Issues

Let,

Page fault service time = 'S'

Main memory access time = 'M'

Page fault rate = 'P' where $0 \leq P \leq 1$

The effective memory access time is formulated as-

$$EMAT = P \times S + (1 - P) \times M$$

(Assume, page table access time is neglected)

8.3.2 Demand Paging with TLB

Assume, TLB hit ratio = 'h'

TLB access time = 'c'

Page fault service time = 'S'

Main memory access time = 'M'

Page fault rate = 'P' where $0 \leq P \leq 1$

The effective memory access time is formulated as-

$$EMAT = h(c + m) + (1 - h)(c + (P \times S + (1 - P) \times M))$$

8.3.3 Page Replacement Algorithms

FIFO

- When a page fault occurs and all the memory frames are full, FIFO algorithm replaces the oldest page to allocate the page referred by the CPU.
- It is implemented using a queue or time-stamp on pages.
- Sometimes, even on increasing the number of frames, the page fault rate increases. This situation is called Belady's Anomaly.

LRU

- LRU algorithm replaces the page that has not been used for the longest period (least recently used page).
- It is implemented using a stack or counter.

Optimal

- Optimal algorithm replaces the page that will not be used for the longest period in future.
- For a fixed number of frames, optimal algorithm gives the least page fault rate.
- It cannot be implemented in real time as it requires future references.

Note

Reference String is a set of successively unique pages referred in a given list of virtual addresses.

8.3.4 Thrashing

When CPU utilization is low, the OS may increase the degree of multiprogramming. After a certain point, the throughput of the system will gradually decrease as the system spends more time in page replacements owing to the lack of frames. This phenomenon is called 'thrashing'.

8.3.5 Working Set Model

It is defined as the set of the unique pages referred during the past ' Δ ' references.

- If $\Delta < (\text{total number of frames})$, the OS can bring in more processes.
- If $\Delta > (\text{total number of frames})$, the OS should instruct the mid-term scheduler to suspend some of the processes to avoid thrashing.

